Materials Characterization

Date: March 2026

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Interfacial microstructure and strengthening mechanisms of Mo-Re alloy/AlN ceramic brazed joints

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10.1016/j.matchar.2026.116108

Highlights

- The Mo-Ni-Si ternary phases were synthesized in situ within the Mo-Re alloy and AlN ceramic joint for the first time.
- Through characterization and thermodynamic calculation, the reaction and diffusion pathways of Mo-Re in molten Ni-Si filler were revealed.
- The product TiN and AlN ceramics form a semi-coherent interface to improve the interface stability.
- The prepared joints exhibit the highest shear strength value reported so far that can meet the requirements of 1000 °C high-temperature service.

Outline

Abstract

Exploring ceramic/metal joints for reliable service at high temperature is a key issue to broaden the boundary of extreme applications. For the first time, the reliable brazing of AlN and Mo-Re was successfully achieved using a Ni-based MPEA filler in this work. The effects of brazing parameters on the microstructure evolution and mechanical properties of the joints were investigated. The joints brazed at 1150 °C for 15 min exhibited a typical interfacial microstructure composed of AlN ceramic/TiN/Ni(s,s)+λ(Mo_2Ni_3Si)+R($Mo_{55}Ni_{42}Si_3$)/λ(Mo_2Ni_3Si)/δ(Mo_7Ni_7)+λ(Mo_2Ni_3Si)/Mo-Re alloy. The synergistic effect of (i) the in-situ generated Mo-Ni-Si ternary phase and the Ni-based solid solution as matrix, and (ii) the semi-coherent interface between TiN and AlN, improved the bonding strength and high-temperature service characteristics of AlN/Mo-Re joint. The maximum shear strength of the joint reached 129 MPa at room-temperature and 78 MPa at 1000 °C high-temperature, which is the maximum value that has been reported to meet the high-temperature service. This work provides a novel method for joining AlN ceramic and Mo-Re alloy, demonstrating potential for applications in extreme-environment electronic packaging.

Keywords

AlN; Molybdenum-rhenium alloy; Brazing; Mechanical properties; Thermodynamic calculation

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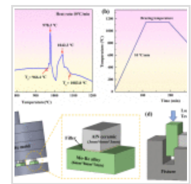
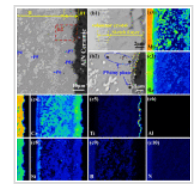
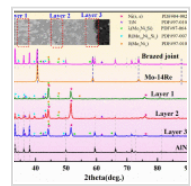
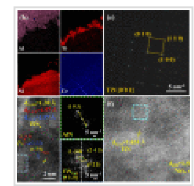
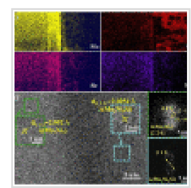
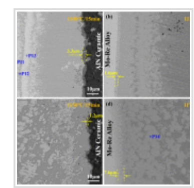
1. Introduction

Aluminum nitride (AlN) metallized substrates have significant potential for wide application in electronic packaging due to their high thermal conductivity and unique

Figures (12)

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Tables (3)

electrical insulation properties [1,2]. Although they are often manufactured in combination with Ag or Cu, the components cannot meet the service requirements of extreme environments, such as high-temperatures exceeding 1000 °C [3,4]. These application shortcomings highlight the importance of AlN ceramic/metal high-temperature heterogeneous connections. Among them, molybdenum-rhenium alloy (Mo-Re) has emerged as an excellent candidate for the manufacture of electronic systems in extreme environments due to its outstanding high-temperature stability [[5], [6], [7]]. The effective connection of AlN and Mo-Re alloy can fully leverage their performance advantages, thereby meeting the dual requirements of insulation and heat dissipation in extreme environment packaging. With the increasing demand for high reliability under harsh conditions, the importance of this integration scheme is becoming increasingly prominent.

Selecting appropriate filler metals and process parameters remains challenging to ensure complete metallurgical reactions during welding and satisfactory high-temperature performance. Notably, there are few reports available in the open literature on the brazing of Mo-Re alloy and AlN ceramic. As for AlN ceramic/metal heterogeneous joints, disparities in physical properties between the substrates, thermal mismatch, and stringent high-temperature service demands hinder improvements in joint mechanical properties [8,9]. Incorporating materials with a low thermal expansion coefficient (CTE) [[9], [10], [11]] or high modulus [12,13] into the filler to form composite brazing alloys is generally an effective approach. Lv et al. [9] successfully realized reliable bonding between AlN and Cu through the introduction of TiN particles into AgCuTi filler, and the maximum joint strength was improved by 150% (131 MPa). TiN not only alleviates the thermal mismatch at the joint interface but also inhibits the formation of brittle intermetallic compounds. Li et al. [13] fabricated an AlN/Cu heterogeneous joint by incorporating $AlCoCrFeNi_{2.1}$ into Ag-based filler. The homogeneous interfacial phases and high bonding strength of the obtained joint were attributed to the FCC solid solution matrix and dispersion-strengthened particles, which collectively alleviated residual stress and enhanced shear strength [14,15]. Nevertheless, Ag-Cu eutectic fillers typically exhibit significant strength degradation and creep at temperatures exceeding 500°C due to the inherent low melting point of Ag-based solid solution [16]. This softening of Ag-based fillers renders such joints unsuitable under elevated temperatures [17]. Evidently, research on high-temperature filler alloys for joining Mo-Re to AlN is still insufficient.

Recently, Mo-Ni-Si ternary metal silicides (such as Mo_2Ni_3Si) have attracted considerable attention due to their ability to form 'in-situ' composites with nickel-based solid solutions, which exhibited higher toughness and improved mechanical properties [[18], [19], [20]]. Inspired by such composites, a Ni-based multi-principal element alloy (MPEA) filler was proposed in this work to connect the Mo-Re alloy and AlN ceramic, with Mo-Ni-Si ternary compounds generated in situ to meet the high-temperature applications. The microstructural evolution and mechanical properties of the joints were analyzed, and the interfacial reaction in AlN/Mo-Re joints was further investigated through thermodynamic calculations.

2. Materials and methods

AlN ceramic was supplied by Suzhou Kematek, Inc. The AlN blocks were cut to $6\times3\times3\,mm^3$ for the bonding experiment. The metallic substrate to be bonded with AlN was the Mo-Re alloy with a chemical composition of Mo~14wt%Re, which was machined to dimensions of $6\times8\times3\,mm^3$. Before brazing, the faying surfaces were first ground and polished with 0.5 μm diamond paste, subsequently ultrasonically cleaned in acetone for 20 min, and then blow-dried. The MPEA filler employed in this study was prepared by ball-milling alloy powders with a composition of Ni~7.2wt%Cr~4.8wt%Ti~0.95wt%Fe~4.2wt%Si~0.5wt%B. Its design was based on the following considerations: (i) NiCr-based alloys have long been utilized as high-temperature filler alloys [21,22], such as Ni-Cr-Si filler alloy for joining molybdenum alloys [23,24]. (ii) Ti plays an active role in improving wettability on the ceramic side [9], and a small amount of Fe can enhance the structural stability of brazing filler metal [25]. (iii) Si and B act as melting point depressants to lower the brazing temperature, while promoting the formation of Mo-Ni-Si ternary phase [17,26,27] in joints, thereby improving the high-temperature service performance. Fig.1a shows the DSC thermogram of the filler, with a melting point of approximately 1083.8°C. The content of each constituent element was determined via preliminary experiments, aiming to ensure that the melting point of the brazing filler metal (1083.8 °C) is consistent with the brazing temperature and to facilitate the in-situ formation of the Mo-Ni-Si phase Subsequently, the MPEA filler was mixed with terpineol to form a paste, which was then applied to the faying surfaces via screen printing. The brazing process was conducted in a high-vacuum resistance furnace with a background pressure of 2.7×10^{-5}Torr, over a temperature range of 1100 to 1175°C. The assembly of the graphite mold and sandwich faying structure is illustrated in Fig.1c. The thermal cycle curve of the brazing process is presented in Fig.1b.

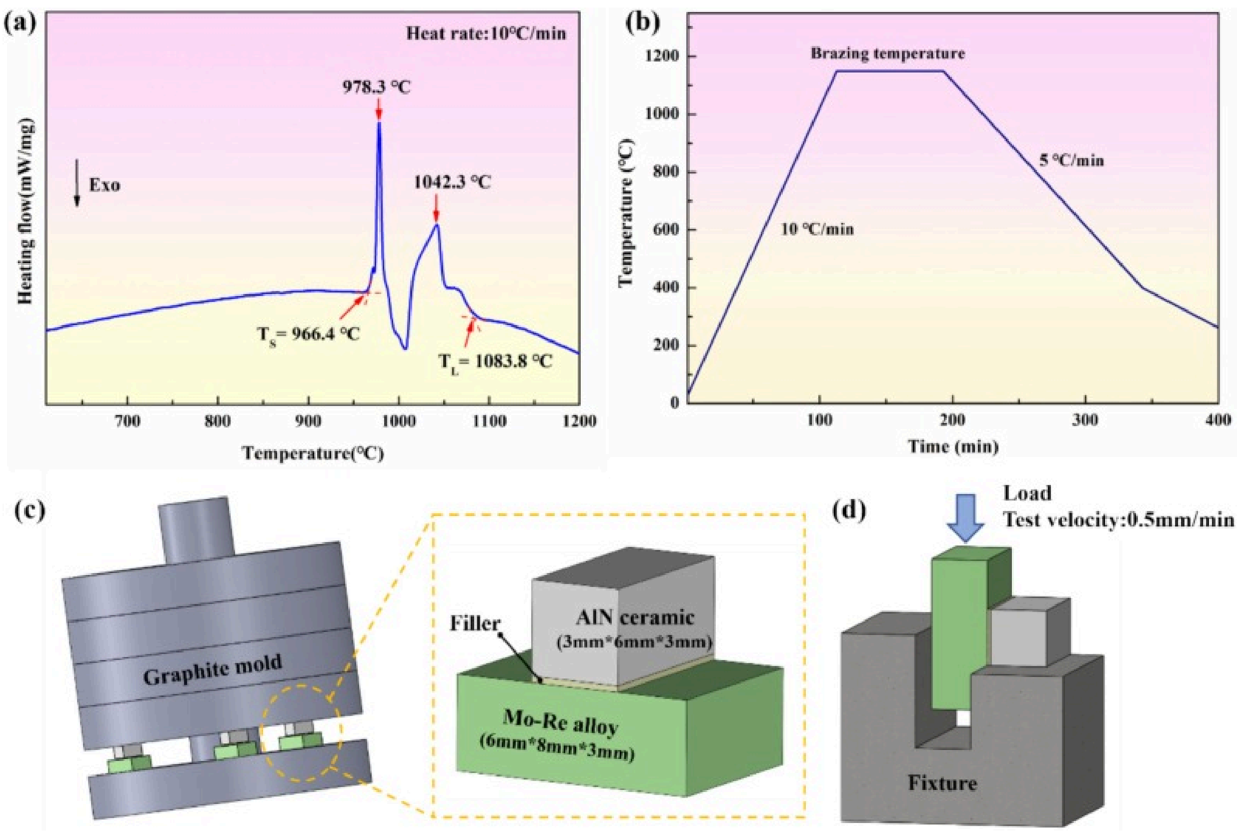

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Fig. 1. (a) DSC thermogram of the MPEA filler; (b) thermal cycle curve of the brazing process; (c) schematic illustration of the assembled specimens; (d) schematic diagram of the shear testing.

In terms of experimental analysis, the microstructure and the chemical compositions of various phases were characterized using an electron probe microanalyzer (EPMA, Superprobe JXA-8230) equipped with wavelength-dispersive spectroscopy (WDS, JXA-8230). The phase composition and reactions of the brazed joints were further characterized using an X-ray diffractometer (XRD, Philips X'pert) with Cu Kα radiation and transmission electron microscopy (TEM, Thermo Fisher Talos F200). The shear properties of the brazed joints at room temperature and 1000°C were evaluated by a universal testing machine (AGXplus 5kN), with three joints per group taken to ensure statistical reliability. Additionally, the elastic modulus and hardness of the interfacial products in the joints were tested using a nanoindentation instrument (Agilent G200) with an applied constant displacement of 500nm and a dwell time of 10s at the maximum load.

3. Results and discussion

3.1. Typical interfacial microstructure of AlN/Mo-Re brazed joint

A typical microstructure of the AlN/Mo-Re joint obtained using the NiCrTiFeSiB MPEA filler at 1150°C for 15min is presented in Fig.2. The corresponding EDS analysis results are listed in Table 1. The filler demonstrates excellent wettability and metallurgical bonding at the interface of substrates, and a continuous and dense reaction layer is clearly visible, as shown in Fig.2a. WDS elemental mapping in Fig.2c reveals a distinct enrichment of Ti near the AlN side, which signifies a strong chemical affinity between Ti and AlN that induces ceramic decomposition. At the same time, the enrichment of the Si element was observed on the Mo-Re alloy side. Meanwhile, the Mo and Re elements showed significant diffusion and dissolution into the brazing seam, as evidenced by the distribution of bright-gray phase.

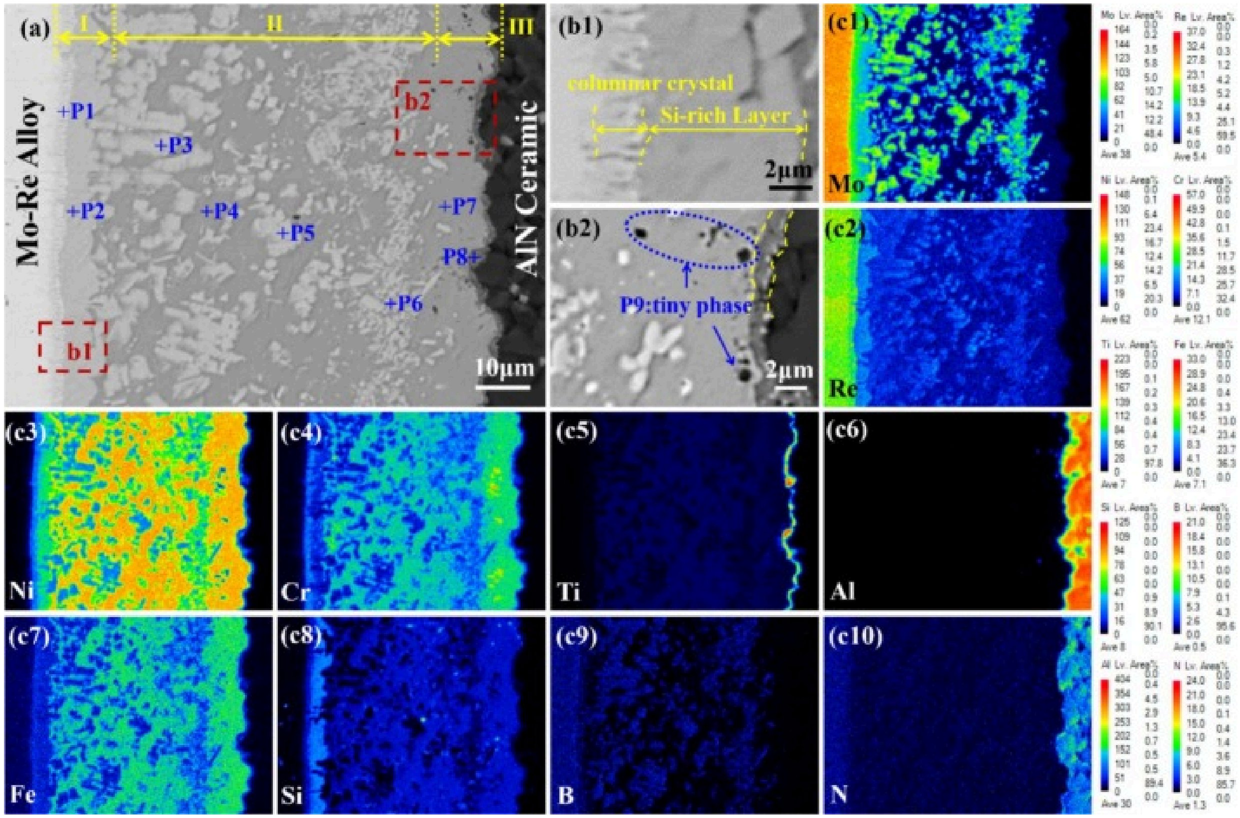

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Fig. 2. Microstructure and elemental distribution of AlN/Mo-Re joint brazed at 1150°C/15 min: (a) overall view of the brazing seam; (b1)-(b2) magnified images of region I and region III; (c1)-(c10) distributions of different elements (WDS).

Table 1. EDS chemical composition in Fig.2 (at.%).

	B	N	Al	Si	Mo	Ti	Cr	Fe	Ni	Re	Possible phase
P1	0.00	0.00	0.83	7.97	45.23	0.91	7.17	1.37	29.66	6.86	δ phase (Mo_7Ni_7) +λ phase (Mo_2Ni_3Si)

	B	N	Al	Si	Mo	Ti	Cr	Fe	Ni	Re	Possible phase
P2	0.00	0.00	0.69	16.38	28.26	0.80	3.90	1.06	45.00	3.91	λ phase (Mo_2Ni_3Si)
P3	0.00	0.00	0.00	2.43	59.97	1.23	2.15	1.18	25.80	7.24	R phase ($Mo_{55}Ni_{42}Si_3$)
P4	0.00	0.00	0.57	5.21	3.23	4.17	8.11	2.86	74.40	1.45	Ni(s, s)
P5	0.00	0.00	0.00	2.22	46.71	2.53	5.41	2.23	36.98	3.92	R phase ($Mo_{55}Ni_{42}Si_3$)
P6	0.00	0.00	0.35	5.79	39.19	3.13	4.65	1.46	43.58	1.85	λ phase (Mo_2Ni_3Si)
P7	0.00	0.00	2.20	7.57	2.29	3.61	6.58	2.28	74.52	0.95	Ni(s, s)
P8	0.00	24.41	15.64	1.60	0.67	38.09	3.10	0.09	15.89	0.51	TiN
P9	0.00	0.00	32.12	4.47	1.65	4.19	3.69	1.77	51.84	0.27	Al_3Ni_5

According to the difference in microstructure and contrast, the whole brazed joint can be divided into three typical regions: the diffusion zone near the Mo-Re alloy side (Region I), the central joint zone (Region II), and the reaction zone adjacent to the AlN ceramic side (Region III). Fig.2b1 shows the local magnification of Region I, which is mainly composed of closely arranged columnar crystals and Si-rich layers. Among them, the Si-rich layer (P2) is primarily composed of Mo, Ni, and Si. It is considered to be λ(Mo_2Ni_3Si), consistent with the findings reported by Zhao et al. [26]. This ternary silicide, also known as MoNiSi [27,28], exhibits a wide range of chemical composition and excellent hardness and toughness [18]. According to the EDS results and the Mo-Ni-Re ternary phase diagram [29,30], it is inferred that the diffusion zone columnar crystal (P1) is a mixed phase composed of δ(Mo_7Ni_7) and λ(Mo_2Ni_3Si). Driven by the concentration gradients, the mixed-phase columnar crystals grow toward the center of the joint, and tightly arranged columnar crystals are formed at the Mo-Re alloy interface [23].

Within this mixed structure, Re and Si exhibit distinct distribution behaviors. Re exists as a substitutional solid solution in the δ phase. Yaqoob et al. [29] investigated the binary intermetallic compound δ phase in the Mo-Ni system, and observed its significant extension in the Mo-Ni-Re ternary system. The maximum solubility of Re in the δ phase is 23.3 at.% in the δ phase. In contrast, Si tends to form intermetallic compounds rather than remain in the solid solution due to its high electronegativity. In Region I, Si is divided into two forms: a small amount enters the δ phase in the form of a trace solid solution, while most reacts with Mo and Ni to form the λ phase. This λ phase exhibits two distinct microstructural features: fine dispersions within the mixed phase (P1) and an observed Si-rich layer (P2). Notably, the Mo-Ni-Si ternary phase diagram [28] confirms the coexistence of the δ and λ phases in equilibrium, which verifies their stable coexistence in the joint. This equilibrium explains the dual distribution of the λ phase (in P1 and as P2).

In Region II, the main constituents comprise a substantial bright-gray phase alongside a medium-gray matrix phase. The medium-gray matrix phase (P4), primarily composed of Ni, is identified as a Ni-based solid solution (Ni(s, s)) with dissolved Cr and Fe. EDS point scan results are in Table 2, highlighting that the bright-gray phases (P3, P5, and P6) mainly consist of Mo and Ni. Notably, the Mo and Re contents in P3 phase surpass those in P5 and P6, reflecting its proximity to the Mo-Re alloy matrix. Correspondingly, P6 has higher Ni and Si contents. Based on the Mo-Ni-Si ternary phase diagram [28], the P3 and P5 correspond to the R phase ($Mo_{55}Ni_{42}Si_3$), while P6 corresponds to the λ phase, the same as P2. It has been reported to be in equilibrium with these Mo-Ni-Si ternary phases [28].

Table 2. EDS chemical composition in Fig.6 and Fig.7 (at.%).

	B	N	Al	Si	Mo	Ti	Cr	Fe	Ni	Re	Possible phase
P10	0.00	0.00	0.60	15.62	30.8	1.04	2.73	1.29	43.20	4.72	λ phase (Mo_2Ni_3Si)
P11	0.00	0.0	0.53	3.62	48.93	1.65	4.36	1.48	35.75	3.68	R phase ($Mo_{55}Ni_{42}Si_3$)
P12	0.00	0.00	1.23	7.63	4.47	1.45	7.36	2.92	73.54	1.40	Ni(s, s)
P13	0.00	0.00	0.37	2.45	49.21	2.22	5.96	1.29	34.51	3.99	R phase ($Mo_{55}Ni_{42}Si_3$)
P14	0.00	0.00	1.86	4.69	26.31	1.52	6.78	2.46	53.85	2.53	λ phase (Mo_2Ni_3Si)

	B	N	Al	Si	Mo	Ti	Cr	Fe	Ni	Re	Possible phase
P15	0.00	28.31	10.64	1.62	20.63	12.25	3.20	0.97	18.89	3.49	MN
P16	0.00	0.00	0.83	2.39	38.67	2.42	8.13	1.58	42.15	3.83	λ phase (Mo_2Ni_3Si)

Fig.2b2 shows a magnified view of Region III, which consists of a medium-gray matrix, a gray-black reaction layer, and trace amounts of tiny black phases. The composition of the medium-gray matrix is similar to that of the matrix phase in Region II, confirming that it is the same Ni(s, s). Consistent with the elemental mapping in Fig.2c5, which shows Ti enrichment near the AlN side, the continuous gray-black phase (P8) adjacent to AlN is primarily composed of Ti and N, and is preliminarily speculated to be TiN. Additionally, tiny black phases (P9) are found near TiN. Their main elements are Ni and Al with an approximate ratio of 5:3, indicating that this phase is Al_3Ni_5.

Subsequently, micro-XRD analyzes were performed on each distinct region of the joint using a layer-by-layer grinding technique to achieve micrometer-scale spatial resolution, as shown in Fig. 3. It can be confirmed that TiN is formed on the AlN surface during brazing. Meanwhile, the diffraction peaks of the λ, δ, and R phases, which are the products of the reaction between Ni, Si, and the elements on the Mo-Re side, essentially correspond to the observations in Table 1.

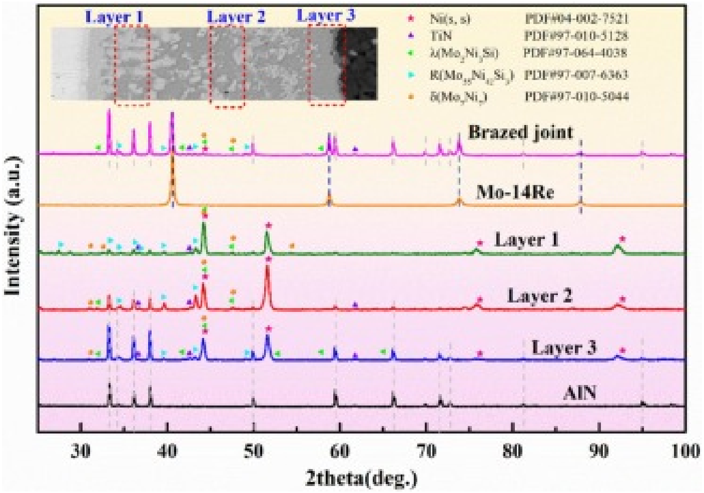

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Fig. 3. XRD patterns of the AlN/Mo-Re joint.

To further confirm the interfacial reaction products identified in the above analysis, TEM characterization was performed on Region I and Region III, as shown in Fig.4 and Fig.5. As shown in Fig.4, a continuous and dense Ti-rich interface reaction layer is formed between AlN and the seam. The selective area electron diffraction (SAED) pattern (Fig.4c) confirms its corresponding TiN, which is a typical interface product formed by the reaction of the active element Ti in the filler with the AlN ceramic. Analysis of the HRTEM image of the AlN/TiN interface (Fig.4e) reveals a specific orientation relationship between the two. The lattice mismatch between AlN and TiN is 3.0%, which belongs to the semi-coherent interface range, due to the ordered arrangement of atoms at the interface, and the interface forms a good combination. Additionally, the SAED pattern of the brazing seam confirms the existence of Ni(s, s). Quantitatively assessed using the formula $\delta = 2\left|\left(\mathrm{d_{TiN}} - \mathrm{d_{Ni(s,s)}}\right)/\left(\mathrm{d_{TiN}} + \mathrm{d_{Ni(s,s)}}\right)\right|$ [26], the lattice mismatch between TiN-Ni(s, s) interface is 15.8%, indicating that the interface is incoherent. Despite the relatively large lattice mismatch, the Ni(s, s) with FCC structure exhibits excellent plastic deformation capacity. It can effectively accommodate interfacial strain caused by lattice mismatch through plastic deformation, thereby reducing residual stress and improving joint strength.

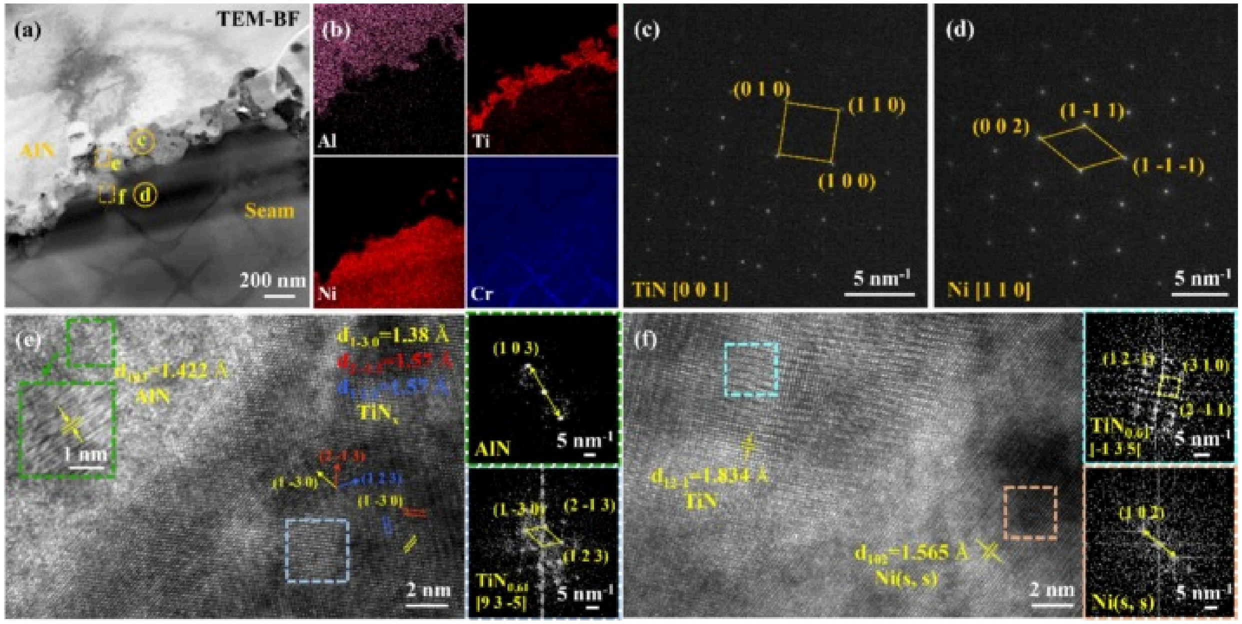

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Fig. 4. TEM analysis of the reaction layer on the AlN side: (a) bright field image of the interface; (b) elemental mapping; (c)-(d) SAED pattern of interfacial product marked in result of Fig.4a; (e) HRTEM image of the interface between AlN and TiN; (f) HRTEM image of the interface between TiN and Ni(s, s).

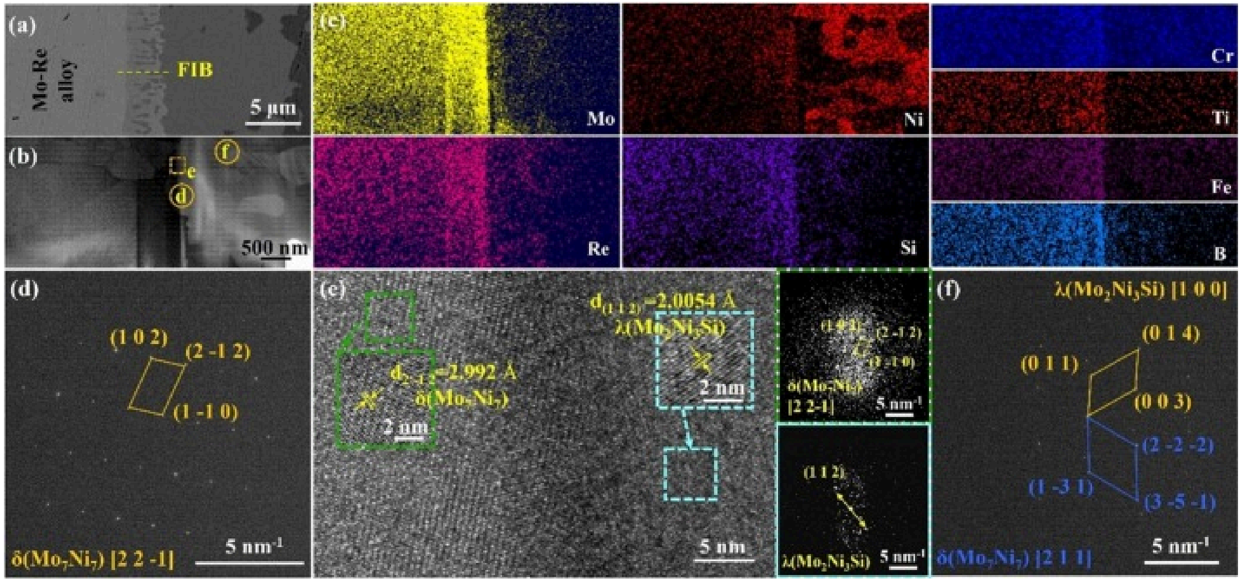

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Fig. 5. TEM analysis of the reaction layer on the Mo-Re side: (a) FIB cutting sampling position; (b) bright field image of the interface; (c) elemental mapping; (d) SAED pattern of interfacial product marked in result of Fig.5b; (e) the HRTEM image of the interface between λ(Mo_2Ni_3Si) and δ(Mo_7Ni_7); (f) SAED pattern of interfacial product marked in result of Fig.5b.

The TEM results of the Mo-Re substrate/seam interface are presented in Fig.5. It can be observed that a silicon-rich layer with a thickness of approximately 500 nm is formed on the Mo-Re substrate, indicating a high chemical affinity between Si and Mo. In addition to the silicon-rich layer, a nanoscale reaction layer with high Ni content was observed, resulting from the synergistic reaction among Ni, Mo, and Si. SAED detected the presence of δ(Mo_7Ni_7) and λ(Mo_2Ni_3Si) phases (Fig.5d and Fig.5f), and the two were closely combined. The orientation relationship between them was [221]||[112]. This ordered orientation relationship can reduce the lattice distortion energy at the interface of the two phases and improve the interfacial bonding stability. Notably, no Mo-Si binary compounds are detected in the entire interface region. Combined with the above phase analysis, it is preliminarily inferred that the chemical affinity between Si and Mo is not directly reflected by the Mo-Si reaction, but transferred through δ(Mo_7Ni_7) acting as a diffusion medium for Si. This speculation will be discussed in Section 3.4 using thermodynamic calculations.

Thus, it can be preliminarily determined that the typical interfacial microstructure of the joint brazed with NiCrTiFeSiB MPEA filler at 1150 °C/15 min is Mo-Re alloy/ δ(Mo_7Ni_7)+λ(Mo_2Ni_3Si) /λ(Mo_2Ni_3Si)/ Ni(s, s)+λ(Mo_2Ni_3Si)+R($Mo_{55}Ni_{42}Si_3$) /TiN/AlN ceramic.

3.2. Effect of the thermal activation on the microstructure of AlN/Mo-Re joints

Fig.6 presents the interfacial microstructure of AlN/Mo-Re joints brazed with MPEA filler for 15 min at different brazing temperatures (1100 °C–1175 °C). As the brazing temperature increases, enhanced reactivity between the substrates and filler not only induces progressive seam widening but also alters the morphological distribution of phases within the joints.

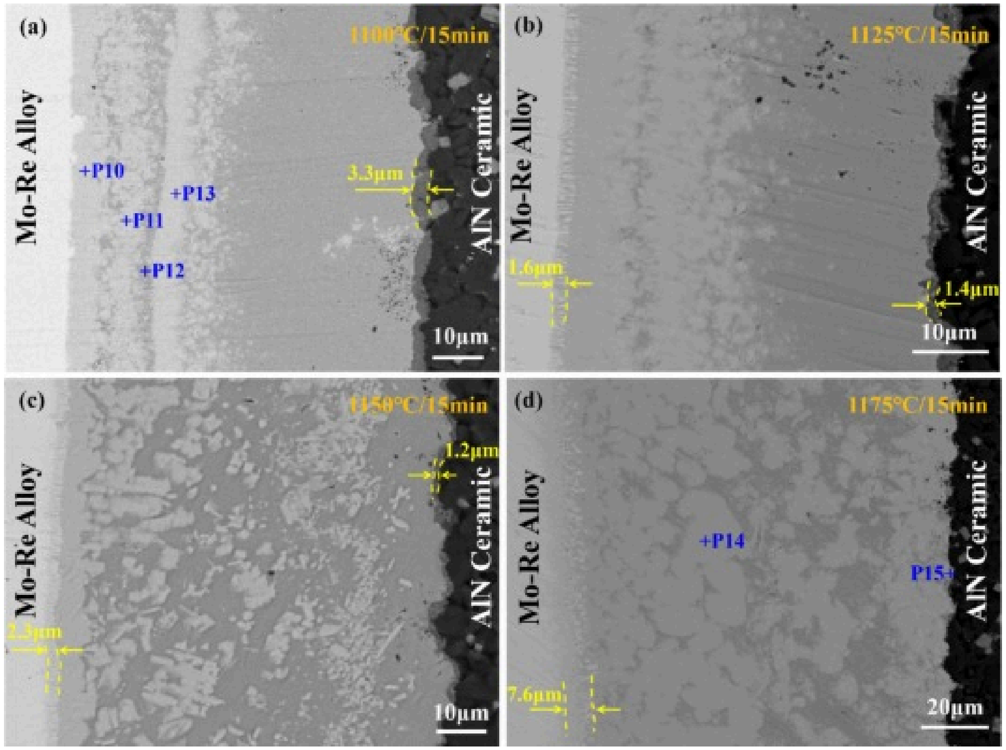

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Fig. 6. Interfacial microstructure of joints brazed at distinct temperatures for 15 min. (a) 1100 °C; (b) 1125 °C; (c) 1150 °C; (d) 1175 °C.

Specifically, at the lower temperature of 1100 °C, limited interfacial reactivity and restricted elemental diffusion from the base materials resulted in narrow joints. Columnar crystals adjacent to the Mo-Re alloy were barely visible, replaced by layered aggregations of the λ and R Mo-Ni-Si ternary phases (P10, P11, and P13). Notably, these layered aggregations are separated by a thin continuous Ni(s, s) layer (P12), which Wang et al. [31] associated with the phase transition of MoNiSi. When the temperature was further elevated to 1125 °C, columnar crystals progressively formed and grew in width, reaching approximately 1.6 μm. Meanwhile, the number of reaction products in the seam increased significantly, particularly the proportions of bright-gray Mo-Ni-Si ternary phases. At 1175 °C, the columnar crystals widened to 7.2 μm, and the boundary between Region II and Region III became blurred, and numerous agglomerated bright-gray λ phases (P14) form in the joint and diffuse toward the AlN side. At this parameter, the microstructure of the central seam region exhibited a high degree of similarity to the γ-Ni/Mo_2Ni_3Si in-situ composite fabricated by Xu et al. [32] through laser melting, in which the dendritic phase corresponds to the ternary metal silicide λ, whereas the interdendritic regions consist of Ni(s, s). This was attributed to the intensified dissolution of the Mo-Re alloy at higher temperatures.

Concurrently, the thickness of the TiN reaction layer decreased significantly. It evolved from 3.3 μm at the initial temperature of 1100 °C to 1.4 μm at 1125 °C, 1.2 μm at 1150 °C, and to an almost invisible thickness at 1175 °C. The trend in the thickness of the TiN reaction layer is related to element competition at the interface. As the brazing temperature increases with a constant holding time, accelerated dissolution of the Mo-Re alloy causes some Mo to dissolve into the liquid phase and diffuse toward the ceramic side. Similar to Ti, Mo acts as an active element and reacts with the ceramic to form a metallic nitride MN (P15) on the ceramic side. This competition between Mo and Ti primarily drives the thinning and eventual disappearance of the TiN layer.

Notably, the microstructure distributions of joints obtained at 1125 °C and 1150 °C differ considerably. Since temperature and holding time are known to regulate the microstructure of brazed joints by controlling thermal activation [33], joints brazed at 1125 °C with varying holding times (15–60 min) were examined, as shown in Fig.7. Similar to the effect of increasing temperature, the seam structure evolved from lamellar reaction layers to agglomerated bright-gray phases with prolonged holding time.

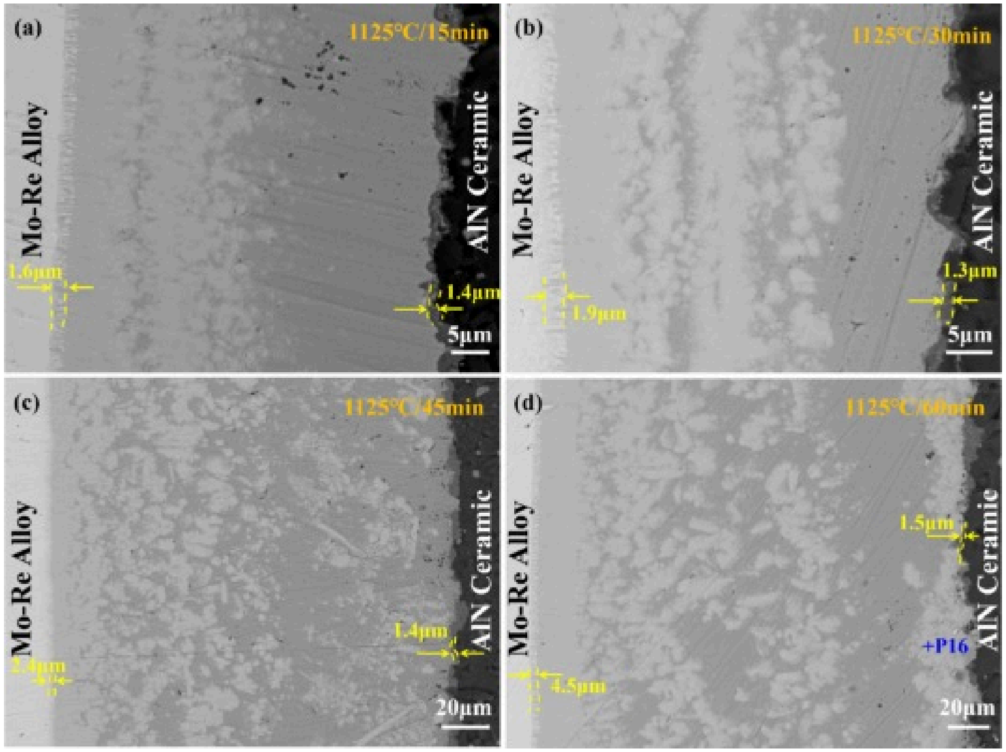

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Fig. 7. Interfacial microstructure of joints brazed at distinct holding times at 1125 °C. (a) 15 min; (b) 30 min; (c) 45 min; (d) 60 min.

In contrast, both the columnar crystal layer on the Mo-Re alloy side and the TiN reaction layer on the AlN ceramic side thickened with increasing holding time. For the columnar crystal layer, the thickness evolved from 1.6 μm at an initial holding time of 15 min to 1.9 μm at 30 min, 2.4 μm at 45 min, and finally to 4.5 μm for 60 min. The thickness of the TiN reaction layer increased slightly, rising from 1.4 μm at 15 min to 1.5 μm at 60 min.

Compared with the brazed joint obtained with 1175 °C/15 min brazing parameters, the TiN reaction layer in the joint brazed at 1125 °C/60 min did not disappear completely.

Meanwhile, as Mo diffused toward the ceramic side, the bright-gray λ phase (P16) formed there. This indirectly confirms that, relative to extending the holding time, brazing temperature exerts a more pronounced effect in promoting interfacial reaction competition and altering phase stability.

3.3. Mechanical performance of the AlN/Mo-Re joint

Fig. 8 illustrates the shear strength of the dissimilar joint brazed under different brazing parameters. Either the room-temperature or the high-temperature test exhibits a trend of first increasing and then decreasing with increasing brazing temperature. The maximum values were achieved at 1150 °C, reaching 129 MPa at room temperature and 78 MPa at 1000 °C. Notably, the mechanical properties of the AlN/Mo-Re joint exhibit a degradation at 1000 °C compared to room temperature. This is primarily attributed to several factors: (i) The yield strength of the Ni(s, s) matrix decreases at elevated temperatures, exhibiting a degree of softening. (ii) Higher temperatures increase atomic mobility, which can lead to microstructural instability over prolonged periods, which may contribute to a reduction in strength.

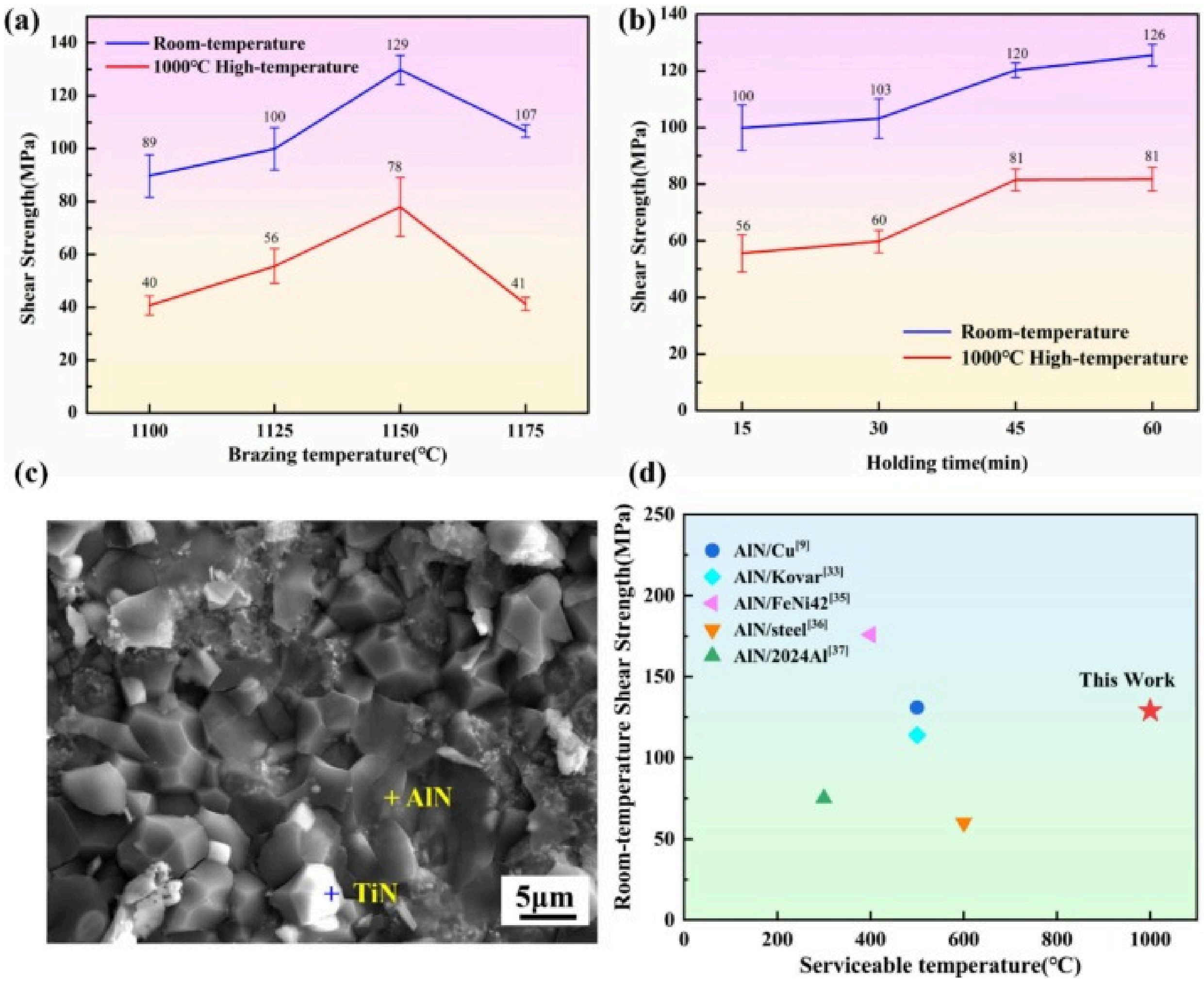

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Fig. 8. Room-temperature and 1000 °C High-temperature shear strength of the joints. (a) brazed at varying temperatures; (b) brazed at varying holding times; (c) fracture surface characterization of AlN/Mo-Re joints; (d) comparation the shear strength of AlN/metal joints reported in the existing literature.

Despite this degradation, the mechanical properties of the AlN/Mo-Re joint exceed the previously reported strength of the AlN/metal joint shown in Table 3, highlighting the excellent bonding ability of the Ni-based MPEA filler and outstanding high-temperature service characteristics. Moreover, when the holding time is extended at a fixed brazing temperature of 1125 °C, both the room-temperature and high-temperature shear strengths of the joints initially increase and then stabilize with prolonged holding time. Specifically, when the holding time reaches 60 min, the joint strength approximates that obtained with the brazing parameters of 1150 °C/15 min.

Table 3. The mechanical properties of the AlN/metal joints reported in the existing literature.

Joint	Joining method	Intermediate layer or filler	Shear strength(MPa)	Whether it meets the high temperature service at 1000°C	Ref.
AlN/Cu	Brazing	AgCuTi+TiN	131	No	[9]
AlN/Kovar	Brazing	AgCuTi+Cu	114	No	[34]
AlN/Cu	Laser irradiation	/	/	No	[35]
AlN/FeNi42	Sputtering+brazing	AgCu19.5Ti3In5	176	No	[36]
AlN/steel	Diffusion bonding	Al	50–60	No	[37]
AlN/2024Al	Brazing	Sn9Zn	75	No	[38]

The fracture surfaces shown in Fig.8(c) exhibit distinct traces of AlN cleavage, confirming that the primary failure occurs on the ceramic side and propagates into the TiN reaction layer. The fracture mechanism can be attributed to the residual thermal stress in the joint resulting from the mismatch in CTE between the ceramic and the brazing filler. Notably, the high brittleness and poor plasticity of AlN hinder the effective relaxation of residual stress on the ceramic side [9], which becomes the key factor for crack initiation in the AlN substrate. With increasing brazing temperature or prolonged holding time, the reaction between the filler and the base metal becomes more thorough, enhancing interfacial bonding. Furthermore, the dispersed Mo-Ni-Si phase within the joint effectively alleviates residual stress, thereby improving joint strength.

However, excessive brazing temperatures lead to extensive Mo diffusion to the ceramic side, which will compete with Ti for reactive sites. This competition leads to thinning of the TiN reaction layer, resulting in insufficient interfacial bonding and destroying the CTE gradient at the interface. This synergistic effect renders the AlN/brazing filler interface into a mechanical weak point, ultimately leading to premature failure and a corresponding decrease in joint strength. Specifically, the disruption of the semi-coherent AlN/TiN interface at 1175 °C due to severe Mo competition and MN phase formation compromises the critical CTE buffering effect and interfacial strain accommodation, leading to a sharp decline in strength.

In contrast, for the joint brazed at 1125 °C for 60 min, while Mo diffusion toward the ceramic side does occur and some MN phase (P16) formation is observed, it does not overwhelm the TiN formation or severely disrupt the semi-coherent AlN/TiN interface. The preservation of this stable, semi-coherent interface, along with the continued formation and homogenization of Mo-Ni-Si phases within the seam, collectively maintains a robust CTE gradient and effective stress-relaxation mechanisms. Consequently, the joint strength at 1125 °C/60 min approximates that obtained at the optimal 1150 °C/15 min.

3.4. Interfacial reaction according to the thermodynamics

To analyze the effects of phase evolution and elemental composition on interfacial reactions, thermodynamic calculations were performed based on the Miedema model and the Toop model.

(1) **AlN ceramic/brazing seam interface**

On the ceramic side, the active element Ti in the molten filler metal diffuses into the AlN ceramic and reacts with it to form a TiN reactive layer during brazing. This layer plays a critical role in enhancing ceramic wettability and improving interfacial bonding.

Following this interfacial reaction, the Al element dissolves into the molten filler metal. And due to its rapid diffusion at elevated temperatures, it is redistributed within the brazed seam while further reacting with the filler composition. In the Ni-rich melt, the dissolution enthalpy of Al is −81 kJ/mol, indicating a strong bonding tendency between Ni and Al, which promotes the formation of the Al_3Ni_5 phase. The following reactions occur during this process [9].

$$[\mathbf{Ti}]+\mathbf{AlN} \rightarrow \mathbf{TiN} + [\mathbf{Al}] \tag{1}$$

$$[\mathbf{Al}]+[\mathbf{Ni}] \rightarrow \mathbf{Al_3\,Ni_5} \tag{2}$$

In fact, apart from the active element Ti, Mo (dissolved from the Mo-Re alloy) and Cr (present in the brazing filler metal) may also diffuse toward the AlN side and participate in interfacial reactions. To clarify this, the formation enthalpies of binary systems between these three metallic elements (Ti, Mo, Cr) and N were calculated. The results are presented in Fig.9(a). The standard molar formation enthalpies of Mo-N, Ti-N, and Cr-N compounds are all negative. This indicates that the formation of these metal nitrides is thermodynamically spontaneous. Moreover, the standard molar formation enthalpy of Ti-N is significantly lower than that of the other M-N binary systems, thus reflecting a stronger thermodynamic driving force for Ti-N formation compared to the other.

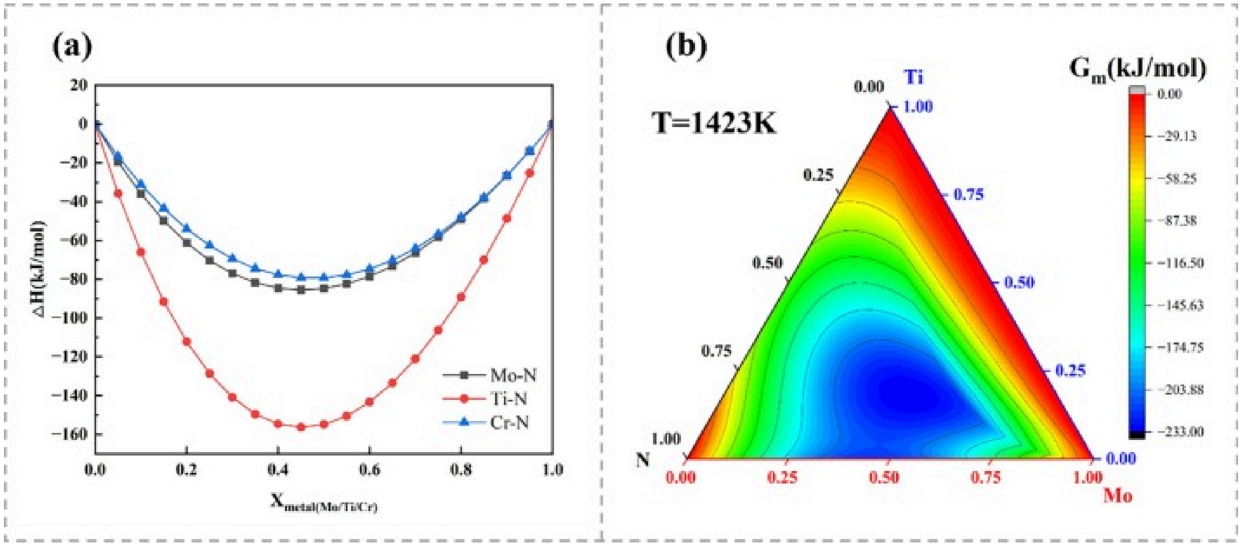

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Fig. 9. Thermodynamic calculation results of (a) formation enthalpy of M-N binary systems; (b) Gibbs free energy of the Mo-Ti-N ternary system.

To further simulate the gradual dissolution and diffusion of Mo from the Mo-Re alloy toward the ceramic side, Gibbs free energy calculations were conducted for the Mo-Ti-N ternary system, with the results shown in Fig.9(b). At lower Mo concentrations, the system tends to form Ti-N, consistent with the observations in Fig.2 and Table 1. With the increase of Mo content, the Gibbs free energy of MoN and multi-component metal nitride (MN) in the Mo-Ti-N system is lower than that of TiN. This aligns with the experimental results in Table 2, which show that the MN formation increases with elevated brazing temperatures or prolonged holding times. It indicates that, under such conditions, active elements other than Ti also participate in interfacial reactions, triggering interfacial reaction competition and thereby promoting the formation of MN. Crucially, the liquid nature of the filler significantly enhances the diffusion kinetics of Mo, allowing it to rapidly transport across the brazing seam and participate in reactions at the ceramic interface, complementing the thermodynamic driving force.

$$[\mathrm{M}]+\mathrm{AlN} \rightarrow \mathrm{MN}+[\mathrm{Al}]\mathrm{M}=\mathrm{Ti}, \mathrm{Mo}, \mathrm{Cr} \tag{3}$$

(2) **Mo-Re alloy/brazing seam interface**

Based on the dominant elements at the Mo-Re alloy/brazing seam interface, the system was simplified to a Mo-Ni-Si ternary system, with trace elements such as Cr, Ti, Fe, and B neglected. As the MPEA filler melts, the Mo-Re alloy comes into contact with the liquid brazing filler, undergoing layer-by-layer dissolution. The following reactions occur during this process [39,40]:

$$\begin{gathered}\mathrm{Mo}+[\mathrm{Ni}] \rightarrow \mathrm{MoNi} \\ \Delta G=-212100+1089T-142T\ln T(\mathrm{kJ}/\mathrm{mol})\end{gathered} \tag{4}$$

$$\begin{gathered}\mathrm{Mo}+[\mathrm{Ni}]+[\mathrm{Si}] \rightarrow \mathrm{Mo}-\mathrm{Ni}-\mathrm{Si} \\ \Delta G=-38980+16.46T(\mathrm{kJ}/\mathrm{mol})\end{gathered} \tag{5}$$

Fig.10 presents the thermodynamic calculation results for the Mo-Ni-Si ternary system at 1150 °C. The negative Gibbs free energies confirm the thermodynamic spontaneity of δ(Mo_7Ni_7) and Mo-Ni-Si phase formation (Fig.10a). Thermodynamic calculations reveal that λ(Mo_2Ni_3Si) exhibits the lowest Gibbs free energy, which is consistent with the experimental observation that the λ phase is enriched at the joint center. For the R($Mo_{55}Ni_{42}Si_3$), owing to its lower Si content, it preferentially forms at the interface of the Mo-Re alloy and gradually transforms into the λ phase with the diffusion of Si.

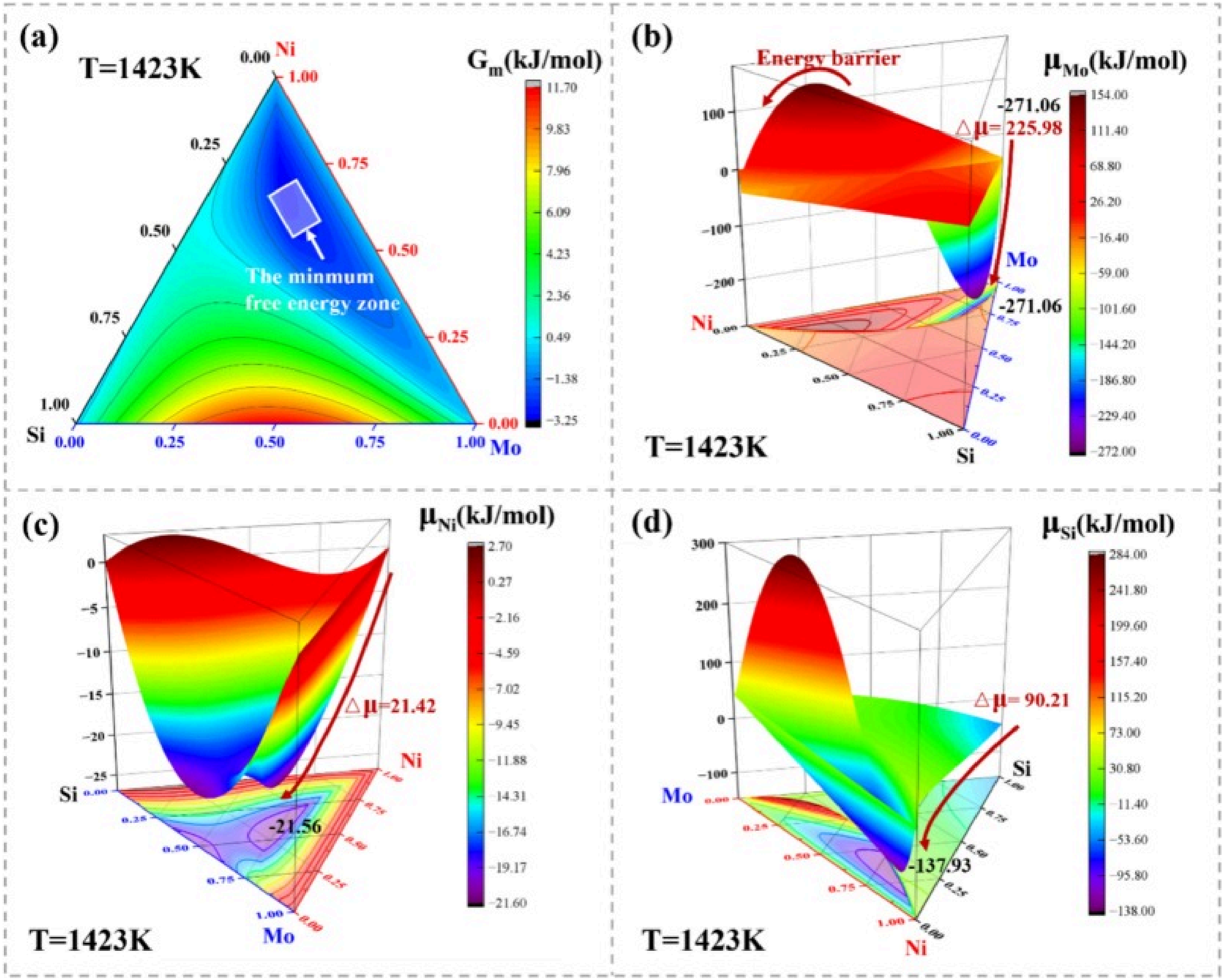

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Fig. 10. Thermodynamic calculation results of (a) Gibbs free energy of the Mo-Ni-Si ternary system; (b)-(d) chemical potential of Mo/Ni/Si in the Mo-Ni-Si system.

Fig.10b shows the chemical potential of Mo in the Mo-Ni-Si system, where Mo exhibits a low chemical potential and encounters a diffusion barrier into either Ni or Si. The chemical potential gradient of Mo toward the Mo-Ni-Si phase region reaches

225.98 kJ/mol, providing a driving force for diffusion into the three-phase coexistence zone. These results suggest that chemical reactions among Mo, Ni, and Si facilitate the spreading of the liquid alloy and the dissolution of the Mo-Re alloy.

Furthermore, differences in silicon content between bright-gray phases suggest that the phase transformation of Mo-Ni-Si phases may proceed inward from the interface, driven primarily by Ni and Si elements. Given the high Ni content in the brazing filler, the reaction products of the dissolved Mo-Re alloy are indicated to be predominantly Si-poor R phases. As Si accumulates in the brazing filler, these phases gradually react to form the Si-rich λ phase. The chemical potentials of Ni and Si in the Mo-Ni-Si system (Fig.10c-d) also indicate a diffusion trend toward the three-phase coexistence region due to the maximum chemical potential gradient. Through directional solidification, MoNiSi phases cluster in the center of the brazed seam, with minor dissolution of Cr, Fe, and B elements.

3.5. Microstructural evolution mechanism

To further clarify the microstructural evolution mechanism of the AlN ceramic/Mo-Re alloy brazed joint, a simplified reaction model was established, as illustrated in Fig.11. The formation process of the AlN/Mo-Re brazed joints can be systematically understood by considering three main stages. And the final morphology within the third stage is further influenced by brazing temperature and holding time.

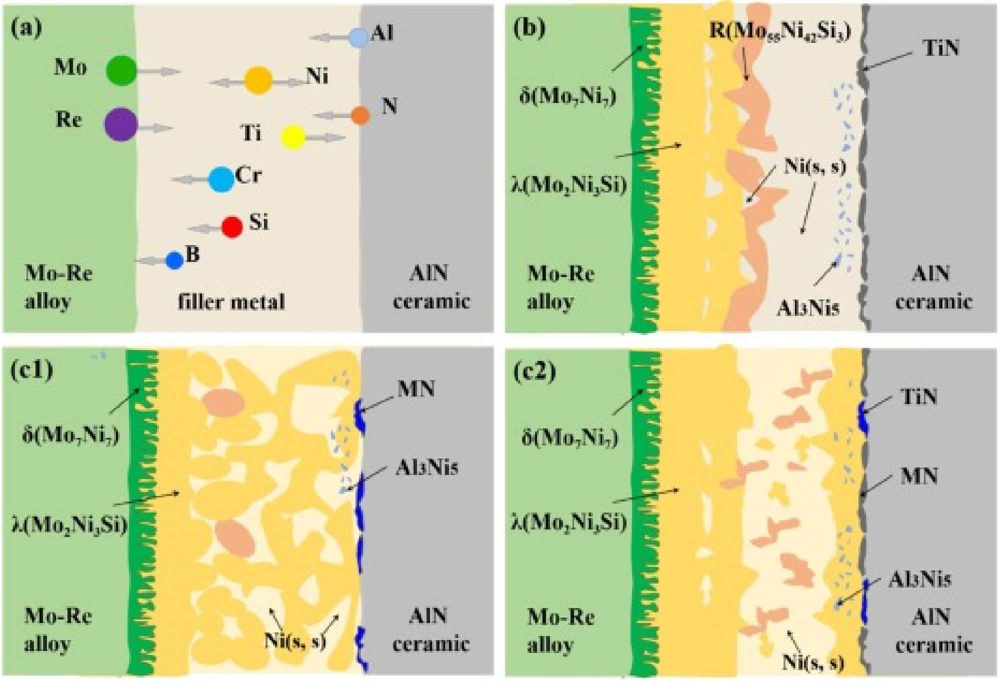

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Fig. 11. Schematic of the joining mechanism of the AlN/Mo-Re brazed joint.
(a) physical contact and atomic diffusion; (b) interfacial reaction and joint formation; (c) Mo-Ni-Si diffusion and homogenization (c1 increase brazing temperature and c2 prolonged holding time)

a. **Physical contact and atomic diffusion stage**

During the heating phase of brazing, a slight pressure is applied to the joint assembly to ensure intimate contact between the filler alloy and both the AlN ceramic and Mo-Re alloy surfaces, facilitating subsequent wetting and interfacial reactions. When the brazing temperature is reached, the filler alloy melts and begins to atomically diffuse widely. The reactive element Ti exhibits strong diffusivity on the surface of AlN. At the same time, the

Mo atoms from the Mo-Re alloy diffuse into the liquid filler driven by the concentration gradient.

b. **Interfacial reaction and joint formation stage**

As Ti diffuses to the surface of AlN, a continuous TiN reaction layer is formed, with minor dispersions of other metal nitrides. Concurrently, the Al atoms released by the reaction between the Ti and AlN diffuse into the liquid filler and react with Ni in the filler to form the Al_3Ni_5 intermetallic phase. In addition, in the diffusion affected zone near the Mo-Re/seam interface, the Mo-Re alloy forms δ(Mo_7Ni_7) containing a significant amount of Re. Within the liquid seam, Mo atoms react with Ni and the low melting point element Si to form two different Si-containing high-temperature Mo-Ni-Si phases, which change the chemical composition of the liquid filler and increase its melting point. In this process, solidification progresses as planar crystals grow from the interface to the center of the seam, and other components (Cr, Fe) are discharged into the residual liquid phase, resulting in their enrichment in the central region.

c. **Mo-Ni-Si phase diffusion and homogenization, influenced by brazing parameters**

With increasing brazing temperature or prolonged holding time, the microstructural evolution within the seam continues, primarily driven by enhanced diffusion and phase transformations.

Effect of Increased Brazing Temperature: At higher temperatures, the strong nitride-forming element Mo continues to diffuse toward the AlN surface, competing more intensely with Ti for nitrogen and promoting the formation of additional MN at the ceramic interface. Concurrently, the dissolution of Mo-Re alloy intensifies, leading to a greater amount of Mo-Ni-Si phases (particularly the λ phase) forming in the central joint region, which often agglomerate and diffuse toward the AlN side.

Effect of Prolonged Holding Time: Extending the holding time at a constant temperature also allows for more thorough diffusion and reaction. On one hand, molybdenum diffuses toward the ceramic surface, forming a small amount of MN phase, yet this does not significantly disrupt the semi-dense AlN/TiN interface. On the other hand, the Mo-Ni-Si phases become more evenly distributed throughout the joint, and the Ni-based solid solution matrix solidifies, forming a ductile matrix with dispersed Mo-Ni-Si phases. This phase equilibrium between the ductile matrix and hard particles further contributes to the overall strength and high-temperature service characteristics.

3.6. Strengthening mechanism of AlN/Mo-Re joints

The brazed joint matrix consists of two distinct Mo-Ni-Si phases dispersed in the nickel-based solid solution. To clearly delineate the mechanical properties of each phase present in the joint, nanohardness tests were conducted, and the results are shown in Fig.12. Firstly, the Ni(s,s) phase is characterized by a low elastic modulus(264GPa) and hardness(7GPa), showing excellent plasticity. In general, these values are significantly lower than those of the Mo-Ni-Si phase, which is conducive to stress relief through deformation-mediated mechanisms.

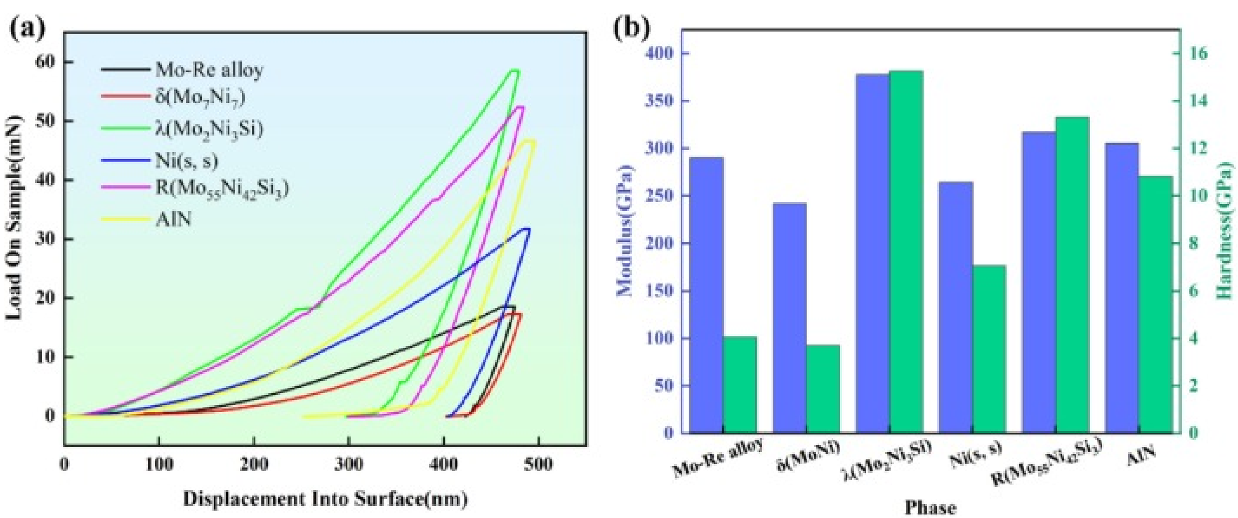

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Fig. 12. Nanoindentation results of the main phases in the AlN/Mo-Re joint. (a) load-displacement curves; (b) the hardness and modulus of different phases.

Secondly, the Mo-Ni-Si phases, characterized by high modulus and hardness, are distributed within the ductile Ni(s,s) matrix, forming a particle-reinforced composite structure at the brazed interface. These mechanical properties contribute to joint strengthening through load-bearing reinforcement, transferring external stress from the matrix to the high-strength particles, thereby enhancing the overall strength.

Thirdly, the semi-coherent interface between TiN and AlN provides a continuous arrangement of atomic layers. The strong covalent bond significantly enhances the anti-separation at the interface ability and alleviates interface strain through dislocation coordination, thereby improving the initial bonding strength of the joint. At the same time, the stability of this interface structure also ensures the joint's high-temperature serviceability.

4. Conclusion

This study successfully achieved reliable joining of AlN ceramic and Mo-Re alloy using a Ni-based MPEA filler, systematically investigating for the first time, the effects of brazing temperature and holding time on the microstructural evolution and mechanical properties of the joints. The following conclusions can be drawn:

(1) During the brazing process, the active element Ti in the liquid solder diffuses and forms a TiN reaction layer with the AlN ceramic. Benefiting from the semi-coherent interface formed by AlN and TiN, the interface strain can be alleviated through dislocation coordination, thereby improving the high-temperature service characteristics of the joint.

(2) Thermodynamic calculation reveals the reaction mechanism between Ni-based multi-component solder and Mo-Re alloy, that is, under the combined action of Ni and Si, Mo-Re alloy dissolves layer by layer, and δ(Mo_7Ni_7) can be used as a diffusion intermediate medium of Si to promote the formation of Si-rich layer λ(Mo_2Ni_3Si) on the surface of Mo-Re matrix.

(3) With increasing temperature or holding time, more Mo-Re alloy dissolves into the weld seam, leading to a gradual increase in the amount of Mo-Ni-Si phases formed via in-situ reactions within the joint. These Mo-Ni-Si phases, when combined with the ductile Ni-based solid solution matrix, form a soft matrix-hard particle composite that effectively enhances the mechanical properties of the joint. A maximum shear strength of 129 MPa at room-temperature and 78 MPa at 1000 °C high-temperature was achieved under the brazing parameter of 1150 °C for 15 min.

CRediT authorship contribution statement

Mengchun Fu: Visualization, Software, Investigation, Writing – original draft. **Panpan Lin:** Supervision, Methodology, Conceptualization, Writing – review & editing. **Yingyao Lu:** Validation, Investigation, Data curation. **Jing Li:** Project administration. **Yang Zhang:** Project administration. **Tiesong Lin:** Supervision, Project administration. **Peng He:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support from “National Natural Science Foundation of China (NSFC, Grant numbers 52175302, U21A20128, U22A20185, and 52305353)”, and “Natural Science Foundation of Heilongjiang Province, China (YQ2024E020)”.

Data availability

Data will be made available on request.

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